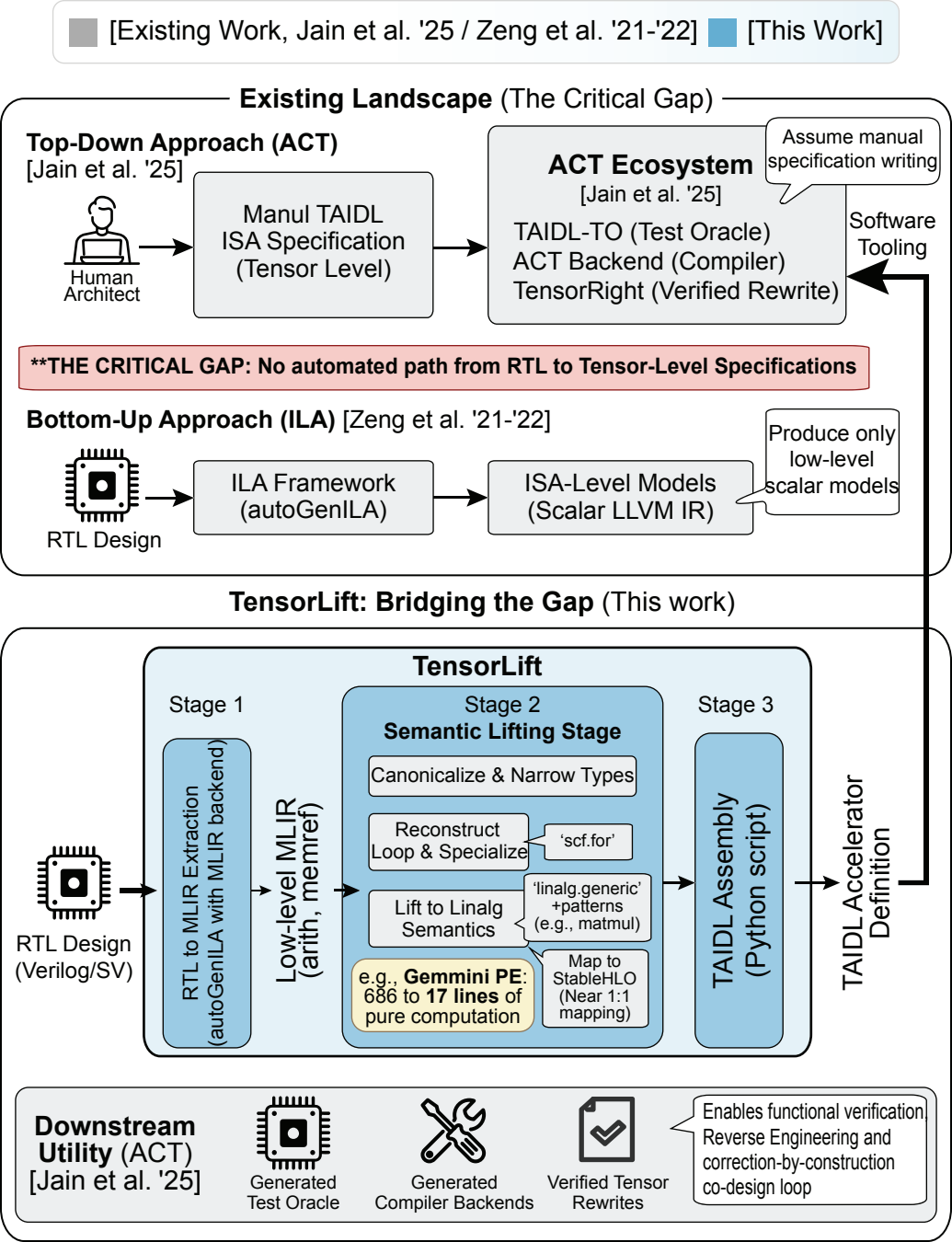
[Existing Work, Jain et al. '25 / Zeng et al. '21-'22]
[This Work]
Existing Landscape (The Critical Gap)
Top-Down Approach (ACT)
[Jain et al. '25]
Human Architect
Manul TAIDL ISA Specification (Tensor Level)
ACT Ecosystem
[Jain et al. '25]
TAIDL-TO (Test Oracle)
ACT Backend (Compiler)
TensorRight (Verified Rewrite)
Assume manual specification writing
Software Tooling
**THE CRITICAL GAP: No automated path from RTL to Tensor-Level Specifications
Bottom-Up Approach (ILA) [Zeng et al. '21-'22]
RTL Design
ILA Framework (autoGenILA)
ISA-Level Models (Scalar LLVM IR)
Produce only low-level scalar models
TensorLift: Bridging the Gap (This work)
TensorLift
Stage 1
Stage 2
Semantic Lifting Stage
Stage 3
RTL Design (Verilog/SV)
RTL to MLIR Extraction (autoGenILA with MLIR backend)
Low-level MLIR (arith, memref)
Canonicalize & Narrow Types
Reconstruct Loop & Specialize
'scf.for'
Lift to Linalg Semantics
'linalg.generic' +patterns (e.g., matmul)
Map to StableHLO (Near 1:1 mapping)
e.g., Gemmini PE: 686 to 17 lines of pure computation
TAIDL Assembly (Python script)
TAIDL Accelerator Definition
Downstream Utility (ACT) [Jain et al. '25]
Generated Test Oracle
Generated Compiler Backends
Verified Tensor Rewrites
Enables functional verification
Reverse Engineering and correction-by-construction co-design loop